Haagerup conformal field theory at central charge eight

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Abstract

We construct a chiral algebra U of central charge eight inside the lattice vertex operator algebra $V_{E_6\oplus A_2}$. It is generated by two commuting affine $\mathfrak{sl}_2$ current algebras at levels 1 and 39, the stress tensor of the lattice theory, and a single weight-two field built from three lattice exponentials. We announce that U is strongly rational and completely unitary, and that $\mathrm{Rep}(U)$ is equivalent, as a unitary ribbon category, to the Drinfeld center $Z(\mathcal{H}_3)$ of the Haagerup fusion category $\mathcal{H}_3$. Its characters are the $\gamma = 0$ Evans–Gannon vector. From U we obtain a diagonal full CFT, and also a holomorphic E_8 extension on which $\mathcal{H}_3$ acts by topological defects.

1 Introduction

Fusion categories describe symmetries of two-dimensional conformal field theories whose topological lines need not be invertible. Many familiar fusion categories arise from finite groups, quantum groups or affine Lie algebras. The Haagerup categories are among the simplest examples outside these standard constructions, so it is natural to ask whether some CFT has Haagerup symmetry. Numerical studies of Haagerup anyon chains and lattice models found evidence for such a CFT at $c \approx 2$ [1, 2]. Here we give an explicit construction of a Haagerup CFT at $c = 8$, starting from its chiral algebra.

Write $\mathcal{H}_3$ for the unitary category with simples $\mathbf{1}, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho$ and fusion rules

$$\alpha^3 = \mathbf{1}, \qquad \rho\alpha = \alpha^2\rho, \qquad \rho^2 = \mathbf{1} + \rho + \alpha\rho + \alpha^2\rho, \qquad d_\rho = \frac{3+\sqrt{13}}{2}. \tag{1}$$

The fusion ring alone does not determine the category: $\mathcal{H}_2$ has the same fusion ring but a different associator, and the two are Morita equivalent [3, 4]. Evans and Gannon studied the modular data of the center and worked out the possible character vectors of a Haagerup VOA at $c = 8$ [5]. More recently, Albert, Honda, Kaidi and Zheng [6] suggested that $(E_8)_1$ has Haagerup symmetry, and that gauging the diagonal Haagerup symmetry of the non-chiral $(E_8)_1$ model gives the $c = 8$ theory predicted by Evans and Gannon. Our approach is to write down a subalgebra of a lattice vertex operator algebra (VOA), so that every product we use is inherited from a fixed, well-understood theory. Our main result is the following.

Theorem 1. *Let U be the algebra defined in (8). With the finite exact lattice inputs specified in Sections 3–5, U is strongly rational and completely unitary, with*

$$c = 8, \qquad U_1 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2, \qquad \mathrm{Rep}(U) \simeq Z(\mathcal{H}_3) \tag{2}$$

as unitary ribbon categories. Its twelve irreducible characters are the $\gamma = 0$ Evans–Gannon vector. There is a diagonal unitary full CFT with $c_L = c_R = 8$. There is also a unitary holomorphic extension $H' \supset U$, isomorphic to V_{E_8}, whose extension algebra $A_3 \in \mathrm{Rep}(U)$ has all-module category $\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3$ and whose algebra-bimodule defects form $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$.

By strong rationality we mean that U is simple, of CFT type, self-contragredient, rational and C_2-cofinite. Complete unitarity means that every simple module carries a positive invariant form and that the canonical fusion forms are positive. In the sketches below we keep apart two kinds of input: a finite number of identities checked by computer algebra, and the infinite-dimensional statements we deduce from them. The computer-algebra inputs are exact identities in lattice coordinates. Matching finitely many character coefficients would not be enough for our purposes. In this note we give the defining fields of U and an outline of the argument identifying the resulting CFT. The full coefficient lists, and records that allow the computations to be replayed, will be made available later with a detailed version.

2 The explicit chiral algebra

Let $L = E_6 \oplus A_2$, with simple-root basis $\alpha_0, \dots, \alpha_7$. The Gram matrix G has diagonal entries 2 and entries -1 on the edges

$$(0,1),\ (1,2),\ (2,3),\ (3,4),\ (2,5),\ (6,7),$$

and zero elsewhere. Set $B = V_L \subset H = V_{E_8}$, using the usual index-three lattice extension, and fix

$$\epsilon(a,b) = (-1)^{\sum_i a_i b_i + \sum_{i>j} a_i b_j G_{ij}}. \tag{3}$$

Coordinates of lattice momenta refer throughout to this basis. The VOA convention is $Y(v,z) = \sum_n v_n z^{-n-1}$ and $T = L(-1)$; in particular

$$Y(e^a, z)e^b = \epsilon(a,b) z^{(a,b)} \exp\left(\sum_{n>0} \frac{a(-n)}{n} z^n\right) e^{a+b}. \tag{4}$$

Together with the Heisenberg relations this fixes the products used below.

Put $\alpha_L = (-1,-2,-3,-2,-1,-2,0,0)$, $c = (5,8,9,8,5,0,2,2)$ and $h_R = \sum_i c_i \alpha_i$. In a rational torus gauge take the currents

$$(H_L, E_L, F_L) = \left(\alpha_L(-1)\mathbf{1}, e^{\alpha_L}, -e^{-\alpha_L}\right), \tag{5}$$

$$(H_R, E_R, F_R) = \left(h_R(-1)\mathbf{1}, \sum\nolimits_{c_i \neq 0} e^{\alpha_i}, -\sum\nolimits_i c_i e^{-\alpha_i}\right). \tag{6}$$

The two triples commute with each other. Their Cartan elements have norms 2 and 78, and $(E_L)_1 F_L = \mathbf{1}$, $(E_R)_1 F_R = 39\mathbf{1}$, so together they generate the simple affine algebra $\mathcal{K} = A_{1,1} \otimes A_{1,39}$, where $A_{1,k} = L_k(\mathfrak{sl}_2)$. The level 39 can be understood as $35 + 4$, coming from the principal embeddings into A_5 and A_2.

Define three norm-four momenta and a field

$$a = \alpha_6 + \alpha_7 - \alpha_5, \qquad b = \alpha_3 + \alpha_4 - \alpha_5, \qquad c' = \alpha_0 + \alpha_1 - \alpha_5,$$
$$u = e^a + \frac{\sqrt{13} + 3\sqrt{-3}}{20} e^b + \frac{\sqrt{13} - 3\sqrt{-3}}{20} e^{c'}. \tag{7}$$

This u is a highest-weight vector of conformal weight two, with Cartan weights $(1,13)$; equivalently, it generates current multiplets of dimensions $(2,14)$. We define

$$\boxed{U = \langle \mathcal{K}, \omega_B, u \rangle \subset B, \qquad \omega_B = \frac{1}{2} \sum_{i,j} (G^{-1})_{ij} \alpha_i(-1)\alpha_j(-1)\mathbf{1}.} \tag{8}$$

We use the stress tensor ω_B of the lattice theory, so $c = 8$. The Sugawara conformal vector of $\mathcal{K}$ has central charge $1 + 117/41 < 8$ and is not the conformal vector used for U.

The rational gauge above is convenient for exact arithmetic, but the positive form has to be transported to it from the standard unitary gauge. Set $(\lambda_0,\dots,\lambda_7)=(5,8,9,8,5,1,2,2)$ and let g multiply e^a by $\prod_i \lambda_i^{-a_i/2}$, fixing oscillators. Transport the lattice Hermitian form and PCT by g. On real oscillator monomials,

$$\theta_g\big(h_1(-n_1)\cdots h_r(-n_r)e^a\big)=(-1)^r\Big(\prod_i \lambda_i^{a_i}\Big)h_1(-n_1)\cdots h_r(-n_r)e^{-a}, \tag{9}$$

antilinearly on coefficients. The form is $\langle v,w\rangle_g=\langle g^{-1}v,g^{-1}w\rangle$. In the original unitary gauge, $g^{-1}u$ is proportional to $e^a+\zeta e^b+\bar\zeta e^{c'}$, with $\zeta=(\sqrt{13}+3i\sqrt{3})/\sqrt{40}$ and $|\zeta|=1$. So the coefficients of u amount to a relative phase, and there is no continuous parameter. The positive real character that defines g extends to E_8, and we use this one unitary structure for the whole chain $U\subset B\subset H$.

3 Finite closure

The first thing to check is that U is really smaller than B: a priori, repeated OPEs of the generators could regenerate the other eighty currents of B, and a computation in the first few weights cannot exclude this. We therefore use a criterion that controls all weights at once.

Set $S=\bigoplus_d S_d$, where $S_d\subset B_d$. The *differential normal span* $\mathcal{N}(S)$ is the smallest linear subspace of B containing S and the vacuum and closed under translation and (-1) products.

Lemma 2 (Recursive strong generation)**.** *Suppose $h\ge 2$ and S_d are finite-dimensional for $1\le d\le h$. Write $K=S_1$ and $A=S_2$. Assume all singular products k_ns, a_ns, for $k\in K$, $a\in A$, $s\in S$ and $n\ge 0$, belong to $\mathcal{N}(S)$. Suppose further that, for $3\le d\le h$,*

$$S_d\subset A_0S_{d-1}+\mathcal{N}_d(S_1+\cdots+S_{d-1}). \tag{10}$$

Then $\mathcal{N}(S)$ is a vertex subalgebra.

Proof sketch. The zero modes a_0 are derivations commuting with T, so the first hypothesis already implies $a_0\mathcal{N}(S)\subset\mathcal{N}(S)$. Induct on the total weight of a pair of generators, and then on its left weight. Replace a higher left generator by a_0b plus a lower normal expression and use

$$(a_0b)_nt=a_0(b_nt)-b_n(a_0t). \tag{11}$$

The first term has smaller input weight. In the second term every composite or derivative in a_0t reduces by the filtered Wick identity; a single generator leaves the same total weight and a smaller left weight. Normal remainders are handled by the same induction using

$$[x_\lambda:yz:]=:[x_\lambda y]z:+:y[x_\lambda z]:+\int_0^\lambda[[x_\lambda y]_\mu z]\,d\mu.$$

The integral has smaller total input weight. All generator singular products therefore close, and Wick induction extends closure to the entire normal span [7]. □

We now describe the input to the lemma. The exact lattice calculation gives a PCT-stable space $S_2\subset U_2$ of dimension 120, generated from the specified currents, ω_B and u by recorded mode recipes. Its current multiplets are

$$(1,5),\ (1,9),\ (1,9),\ (1,13),\ (2,2),\ (2,6),\ (2,8),\ (2,14).$$

They contribute 96 states, in addition to 22 affine vacuum descendants and two neutral fields. The constructed space S_3 has dimension 956, with

$$N_3=K_{-1}S_2+TS_2,\qquad \dim N_3=634,\qquad S_3=N_3+(S_2)_0S_2. \tag{12}$$

Applying current projectors to $(S_2)_0 S_3$ gives a current triplet $P_4 \subset U_4$ of highest weight $(0,2)$. Set

$$N_4 = K_{-1}S_3 + TS_3 + (S_2)_{-1}S_2, \qquad W_4 = N_4 + P_4,$$
$$N_5 = K_{-1}W_4 + TW_4 + (S_2)_{-1}S_3. \tag{13}$$

By construction $P_4 \subset (S_2)_0 S_3 + N_4$. Apart from the singular-product identities in lower weight, the two key checks are

$$(S_2)_0 S_3 \subset W_4, \qquad (S_2)_0 P_4 \subset N_5. \tag{14}$$

We verify these as equalities of vectors over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$, and we record how each field was generated.

To give a sense of the size of the computation: the quotient S_3/N_3 has dimension 322. For the first check, we verify the inclusion on a set of seed vectors. Together with the two commuting current raising operators, these span the quotient. If a quotient M of it satisfies $M = E_L M + E_R M$, then $M = 0$ because the raising operators are jointly nilpotent, so the identities propagate from the seeds. The remaining classes that are singlets under both currents are recovered using exact current projectors and lowering identities that are checked separately. The second check uses the way the triplet was constructed, the Jacobi identities, and an explicit witness for the bottom current.

We now apply Lemma 2 to K, S_2, P_3, P_4, where P_3 is a space of lifts of S_3/N_3 to U_3. The outputs we need have weight at most five. The subalgebra produced by the lemma contains the defining fields of U and is contained in the algebra they generate, so it is U. Hence U is strongly generated in weights at most four, and

$$\dim U_1 = 6, \qquad \dim U_2 = 120, \qquad \dim U_3 = 956. \tag{15}$$

Since the generators are PCT stable, the vacuum form on U is positive and U is simple. In particular $U \neq B$, because $\dim B_1 = 86$.

4 From positive stress identity to rationality

For rationality we need two more pieces of finite input. The first is that U contains copies of the parafermion algebras $K(\mathfrak{sl}_2, 3)$ and $K(\mathfrak{sl}_2, 13)$, which are regular and unitary [8]. Their conformal vectors p and z have central charges $4/5$ and $8/5$, and on their states their own grading and translation agree with those inherited from B. The second is the exact weight-two identity

$$\omega_U = \omega_{A_{1,1}} + \frac{3403}{4563}\omega_{A_{1,39}} + \frac{25}{6}\bar{p} + \frac{25}{26}\bar{z}. \tag{16}$$

A bar denotes averaging over the compact symmetry group generated by the current zero modes. This projects each field onto its invariant component. The group acts unitarily with respect to the inner product $\langle \cdot, \cdot \rangle_g$ defined in Section 2. Note that (16) is not a decomposition of ω_U into commuting pieces. Each average lies in the convex hull of a compact subset of a finite-dimensional space, so it can be replaced by a finite convex combination. We can therefore write $\omega_U = \sum_i t_i e_i$ with all $t_i > 0$, where each e_i is the conformal vector of a regular, unitary subalgebra A_i whose grading is compatible with that of U.

Identifying the second parafermion algebra takes more than matching its central charge. We check that its weight-three primary w, its stress tensor z, and the fields $w_1 w$ and $w_1(w_1 w)$ have exactly the singular OPEs of $K(\mathfrak{sl}_2, 13)$ in the first row, including the simple poles and the decoupling relation at weight six [9, 10]. Iterating and using Wick identities give the remaining OPEs among the generators. Now consider, in the product of the two simple realizations, the subalgebra generated by the diagonal pairs. It is of CFT type, so it has a unique maximal proper graded ideal. The kernels of the two projections therefore coincide, and this gives the identification. The finite inputs to this argument are membership in U, PCT stability, the complete first rows, and (16).

Proposition 3. *The positive identity (16), with these embedded regular factors, implies that U is strongly rational. Every nonvacuum simple module has strictly positive lowest weight.*

Proof sketch. We begin with C_2-cofiniteness. For every simple A_i-module, the quotient by its C_2 subspace is finite-dimensional [11]. Since there are only finitely many simple A_i-modules, the energies occurring in these quotients are bounded by some D_i. By regularity, U decomposes as a direct sum of A_i-modules. If $v \in U_n \cap C_2(U)^\perp$, then v has no component of A_i-energy above D_i, since such components already lie in $C_2(U)$. As the $L^{e_i}(0)$ are self-adjoint, we get

$$n\|v\|^2 = \sum_i t_i \langle L^{e_i}(0)v, v\rangle \le \Big(\sum_i t_i D_i\Big)\|v\|^2. \tag{17}$$

So the C_2 quotient vanishes above a finite weight, and U is C_2-cofinite.

Next, let M be a simple U-module with lowest weight h and top space E. Because the gradings are compatible, E is a sum of complete tops of A_i-modules. Each partial energy $L^{e_i}(0)$ is diagonalizable on E with nonnegative real eigenvalues, so

$$h \dim E = \sum_i t_i \operatorname{Tr}_E L^{e_i}(0) \ge 0. \tag{18}$$

If $h = 0$, then every partial energy vanishes on E. Since the vacuum is the only simple A_i-module of weight zero, each partial translation annihilates E, and hence $L(-1)E = 0$. Any nonzero $v \in E$ then gives a module map $U \to M$, $a \mapsto a_{-1}v$, and so $M \cong U$.

Applying the same trace argument to the generalized weight-zero space of an epimorphism $N \twoheadrightarrow U$, we find a vacuum-like preimage of $\mathbf{1}$, which splits the epimorphism. So the vacuum module is projective. The category of grading-restricted generalized U-modules is a finite braided category with Grothendieck–Verdier duality, by C_2-cofiniteness [12]. For any indecomposable object, a nonzero evaluation onto the unit splits because the unit is projective and simple, so every indecomposable is non-negligible. A finite category has at most exponential growth, so the rigidity theorem of Etingof and Penneys for moderate growth applies [13]. In a rigid category with projective unit every object is projective. Hence the category is semisimple and U is rational. Note that this argument does not assume that the nonvacuum modules are unitary. □

5 Branching determines the boundary fusion ring

To find how B and H decompose as U-modules, we need a uniform bound on the lowest weights of U-modules. This uses one more piece of finite input: a rational lattice–Potts subalgebra $\mathcal{T} \subset U$ of central charge $14/5$, commuting with $\mathrm{PF}_{13} = K(\mathfrak{sl}_2, 13)$. The product $A = \mathcal{T} \otimes \mathrm{PF}_{13}$ has pointed charges $(j,k) \in (\mathbb{Z}/13)^2$ with quadratic form $-(j^2+k^2)/13$. In U_2 there are joint eigenvectors at Cartan weights $(\pm1, \pm5), (0, \pm8)$ with $\mathcal{T}$-energy $14/13$, parafermion energy $12/13$, and zero complementary energy. They lie on the simple-current line $k = -5j$. In the zero-complementary-energy subalgebra relative to A, positivity, nonzero inverse-charge products and associativity imply that the supported invertible constituents have multiplicity one and subgroup support; integrality of the grading makes this subgroup isotropic. Adjoining these vectors to A therefore gives a regular extension inside U,

$$A_L = \bigoplus_{k=-5j} \mathcal{T}_{1,j} \otimes \mathrm{PF}_{0,k} \subset U, \qquad \mathrm{Rep}(A_L) \simeq \mathrm{Fib} \boxtimes \mathrm{SO}(3)_{13}. \tag{19}$$

The extension is free on charges and has fourteen simple modules, denoted $x_a = (1,a)$ and $y_a = (\tau, a)$, $0 \le a \le 6$.

For a U-top E, let x_a, y_a also denote the multiplicities of these complete tops, and put $D = \dim E$. If L, Q, P, Z denote the traces of the left affine energy, $H_R(0)^2$, Potts energy, and parafermion energy, respectively, (16) gives

$$hD = W = L + \frac{83}{6084}Q + \frac{25}{6}P + \frac{25}{26}Z. \tag{20}$$

The dimensions and traces for the fourteen tops are obtained by a finite minimization over charges. For $a = 0, \dots, 6$ they are

a	0	1	2	3	4	5	6
$\dim x_a$	1	1	5	9	2	14	6
$4680W(x_a)$	0	600	10440	56640	5520	148560	54240
$\dim y_a$	1	7	15	2	14	6	2
$4680W(y_a)$	7800	36960	110880	4080	81480	18240	4920

Since the full U-top E carries a complete representation of the right $\mathfrak{sl}_2$, its multiplicities satisfy $y_2 \geq x_5$ and $y_4 \geq x_5 + x_6$. The following exact identity then bounds the trace:

$$\begin{aligned} 4680(\tfrac{17}{10}D - W) = {} & 8460(y_2 - x_5) + 28716(y_4 - x_5 - x_6) \\ & + 7956x_0 + 7356x_1 + 29340x_2 + 14964x_3 + 10392x_4 \\ & + 22212x_6 + 156y_0 + 18732y_1 + 11832y_3 \\ & + 1188y_4 + 29496y_5 + 10992y_6 \geq 0. \end{aligned} \tag{21}$$

So every simple U-module has lowest weight $0 \leq h \leq 17/10 < 2$. Note that the estimate only uses traces on the top; we do not need the partial energies to be simultaneously diagonalizable, nor a positive form on E.

The last finite inputs are that the zero modes act irreducibly on the eighty currents of B outside U, and on each of the four mixed lattice tops of dimension eighty-one. We check this exactly by exhibiting a unique joint eigenline, showing that it spans cyclically, and verifying adjoint identities for the positive metric. Any further summand with integral weights would have lowest weight at least two, which the bound rules out. This gives the decompositions

$$B \downarrow_U = U \oplus X, \qquad H \downarrow_U = \mathbf{1} \oplus X \oplus Y \oplus Y^*, \quad h_X = h_Y = 1, \quad (\dim X_{\text{top}}, \dim Y_{\text{top}}) = (80, 81). \tag{22}$$

Put $\mathcal{C} = \mathrm{Rep}(U)$ and $\mathcal{F} = \mathcal{C}_H$, the category of all right modules over the extension algebra. Holomorphicity makes H Lagrangian, so $\mathcal{C} \simeq Z(\mathcal{F})$ [14]. Since all nonvacuum weights are strictly positive, character asymptotics show that the categorical dimensions are positive, so $\mathcal{F}$ is pseudo-unitary. Induction to the center gives

$$\operatorname{rank} \mathcal{F} = \dim \mathrm{End}_{\mathcal{C}}(H) = \begin{cases} 4, & Y \ncong Y^*, \\ 6, & Y \cong Y^*. \end{cases} \tag{23}$$

The lattice extension H/B gives three invertible objects with trivial associator. Concretely, in the discriminant form $q(a, b) = (2a^2 + b^2)/3$, the isotropic lines (a, a) and $(a, -a)$ are transverse; induction from the second realizes $\mathrm{Vec}_{\mathbb{Z}/3}$ inside $\mathcal{F}$. Moreover X is not invertible. If it were, $B = U \oplus X$ would be a $\mathbb{Z}/2$ extension, and so $E_6 \oplus A_2$ would have an involution whose fixed Lie algebra has rank two. But the fixed rank of any such involution is at least $4 + 1 = 5$.

Suppose first that the rank is four. Then $\mathcal{F}$ is a pseudo-unitary near-group category for $\mathbb{Z}/3$, and the classification leaves the multiplicities $m = 0, 2, 3, 6$ [15]. The case $m = 0$ is the $\mathbb{Z}/2$ extension we have just excluded. When $m = 2$, the classification of [16] reduces us to an A_4 orbifold of V_{E_8}, which has at least $(248 - 3 \cdot 8 - 8 \cdot 4)/12 = 16$ currents. When $m = 6$, the balanced induction pattern would give $N_{XX}^X = 4/3$, which signed Galois symmetry and Verlinde integrality forbid. For $m = 3$ and the other $m = 6$ patterns, we induce to the center and cancel the principal parts. The possible numbers of currents are then

m	$\dim U_1$ allowed by the character projections	
3	$15 + A + 10B$ or $14 + 42A$	$A, B, P \in \mathbb{Z}_{\geq 0}$.
6	$8 + A$ or $5 + 12P$	

None of these equals six. In computing these projections we subtract the prescribed polar terms from modular functions with finite image, and what remains is a constant invariant vector. So

the exclusions apply to the full characters, not only to truncated q-series. So the rank is six. Reading off from $\mathrm{End}(H)$, the complex fusion algebra is $\mathbb{C}\oplus\mathbb{C}\oplus M_2(\mathbb{C})$ [17]. If $\mathcal{F}$ were pointed it would have dimension six; comparing dimensions of the extensions would give $\mathrm{FPdim}\, B = 2$ and $\mathrm{FPdim}\, X = 1$, which we excluded above. Hence there are exactly three invertible objects. Since the fusion is noncommutative, the three noninvertible simples form a single free orbit under the cyclic group, and the right action is by inverses. Frobenius reciprocity then gives

$$\rho^2 = \mathbf{1} + m(\rho + \alpha\rho + \alpha^2\rho), \qquad m \geq 1. \tag{24}$$

It remains to show that $m = 1$, and this is an arithmetic argument. Set $\Delta = 9m^2 + 4$ and let $n_j(\rho + \alpha\rho + \alpha^2\rho)$ be the forgetful images of the residual central simples, with twists θ_j. Induction and the first two Frobenius–Schur indicators [18] give

$$\begin{aligned} \sum_j n_j^2 &= 4m^2 + 2, & \sum_j n_j^2\theta_j &= -\frac{m^2 + m\sqrt{\Delta}}{2}, \\ \sum_j n_j^2\theta_j^2 &= \frac{-m^2 + (2\varepsilon - m)\sqrt{\Delta}}{2}, & \varepsilon &= \nu_2(\rho) \in \{\pm 1\}. \end{aligned} \tag{25}$$

Write $\Delta = t^2 D$ with D squarefree. The root-of-unity length bound [15], applied to the first moment, gives, for $D \neq 10$,

$$4m^2 + 2 \geq \frac{mt}{2}\varphi(2D) \geq \frac{6m\sqrt{9m^2+4}}{\sqrt{13}}.$$

For $m > 1$ this contradicts $36m^2(9m^2+4) - 13(4m^2+2)^2 = 4(m^2-1)(29m^2+13) > 0$.

When $D = 10$ we average over Galois orbits instead. Each unit of the negative rational part then costs at least one root of unity, and each unit of $-\sqrt{10}$ costs at least eight; this excludes $m \geq 4$. The remaining case $m = t = 2$ saturates the bound. Since the second moment also lies in $\mathbb{Q}(\sqrt{10})$, Galois averaging forces its sum of squared roots to be 2, whereas the second moment is -2 or $-2 - 4\sqrt{10}$. Hence $m = 1$. The six remaining twists are then the roots of unity $e^{2\pi i k/13}$ with k a quadratic non-residue mod 13, each occurring once. In particular the center has twelve simple objects.

6 Complete unitarity and categorical identification

So far we know the fusion ring and the modular data. To upgrade this to an equivalence of unitary ribbon categories we pass to conformal nets. The lattice realization gives strong locality and bounded localized vertex operators [19, 20]. By regularity, the subnet generated by bounded insertions has finite index [21], so it is completely rational by Longo's theorem [22], and the comparison with $\mathcal{N}_U$ in [20, Lemma 7.6] transfers complete rationality to $\mathcal{N}_U$. The vacuum representation of $\mathcal{N}_H$ restricts to $\mathcal{N}_U$ as $\mathbf{1} \oplus X \oplus 2Y$: thanks to polynomial energy bounds, the projections onto the algebraic summands reduce the closed smeared fields. Conversely, an intertwiner of the nets commutes with rotations, so on finite-energy vectors it restricts to a module map. Hence multiplicities and inequivalence are preserved. Strong additivity, together with the fact that the vacuum energy space is one-dimensional, shows that $\mathcal{N}_U \subset \mathcal{N}_H$ is locally irreducible, and complete rationality then gives finite index [23, Proposition 2.3].

It follows that the category of boundary sectors of the net inclusion has the same noncommutative rank-six fusion algebra, with a trivially twisted $\mathbb{Z}/3$ subcategory. If it were pointed, then multiplicativity of the index and $[\mathcal{N}_H : \mathcal{N}_B] = 3$ would give $[\mathcal{N}_B : \mathcal{N}_U] = 2$. The corresponding involution lifts to B [19], and the fixed-rank argument above rules this out. With exactly three invertibles, the induction and indicator argument of the previous section goes through and gives twelve sectors of the net. Reconstruction [24, Theorem 9.5 and Remark 9.8] assigns to each of

them a unitary weak U-module. An irreducible sector gives an ordinary simple module. Indeed, by regularity and by working on the common core of bounded-energy vectors, the projection onto the closure of an ordinary summand commutes with the modes, with their closures and adjoints, and hence with the net. So it is the identity. The resulting comparison functor is fully faithful and preserves L_0. The twelve simple sectors of the net therefore account for all twelve simple U-modules, and every simple U-module is unitary.

Positivity of the canonical fusion forms needs one more ingredient: a comparison of annulus insertions. For each localized pair (A, B) of incoming and outgoing annuli and each sector M, we need

$$\pi_M\Big(\overline{B_U Y_U(v,z)A_U}\Big) = \overline{B_M Y_M(v,z)A_M}, \tag{26}$$

where the bar denotes bounded extensions of the sesquilinear insertion forms. We first use the same boundary reparametrization to localize both annulus operators in a common interval. We then transport them to the sector M and undo the reparametrization, using the same choice of implementing operator for both factors [24, Proposition 4.17 and Theorem 9.5]. On finite-energy vectors the formal matrix coefficients agree, by reconstruction and covariance; the inverses of incoming annulus operators are only applied on their dense ranges. Factoring a smaller round annulus through the incoming one provides a dense set of test vectors, and analytic continuation extends the equality over the whole insertion region. In the vacuum sector this places the bounded insertions inside $\mathcal{N}_U(I)$, and Haag duality shows that the net of bounded insertions and the net of smeared fields coincide on the common vacuum Hilbert space. The central factors of the two lifts cancel, including the phases coming from fractional lowest weights, so we obtain (26) without choosing lifts globally. We can now apply Tener's positivity theorem [21, Corollary 4.6] to every unitarizable module, and this proves complete unitarity.

Next we identify the category. The extension H/U now defines a commutative Q-system in $\mathrm{Rep}(U)$, and the category of all right modules over this extension algebra is a unitary fusion category [25]. It has fusion ring (1), and its pointed part has trivial associator. This removes the degree-three obstruction in the classification of generalized Haagerup categories, and the remaining obstruction vanishes because $H^2(\mathbb{Z}/3, \mathbb{C}^\times) = 0$. By Izumi's classification the category is therefore $\mathcal{H}_2$ or $\mathcal{H}_3$ [3, Theorems 5.9 and 9.1]. These two are Morita equivalent [4], so the Lagrangian-center theorem gives $\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3)$. Since the spherical structures are positive, Reutter's uniqueness theorem [26] upgrades this to a unitary ribbon equivalence. We stress that this step relies on the classification of fusion categories, and not on any claim that the S and T matrices determine the category.

7 Characters and the two full theories

Let $q = e^{2\pi i\tau}$ and $\chi_i = \mathrm{Tr}_{M_i} q^{L_0 - 1/3}$. In the Haagerup modular representation the only possible nonvacuum polar term is $\gamma q^{-7/39}$ in the twist-$2/13$ coordinate, with $\gamma \in \mathbb{Z}_{\geq 0}$. By the congruence property the modular representation has finite image [27]. Two character vectors with the same principal part therefore differ by a holomorphic vector of weight zero, that is, by an invariant constant vector. But there is no nonzero such vector: the only coordinate fixed by T has an S-image with nonzero vacuum component. Using the fundamental matrix Ξ of Evans and Gannon, we conclude

$$\boldsymbol{\chi} = \Xi_{:,0} + \gamma \Xi_{:,d_3}, \qquad \dim U_1 = 6 + 13\gamma. \tag{27}$$

Since $\dim U_1 = 6$ by (15), we must have $\gamma = 0$. This fixes every coefficient of every character; for example,

$$\chi_U = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots). \tag{28}$$

We list the twelve sectors below, with $d = (3 + \sqrt{13})/2$ and $s = 1 + d^2 = (13 + 3\sqrt{13})/2$. Note that $\dim M_h$, the dimension of the lowest-weight space, is not the categorical dimension d_i.

Sector	h	$\dim M_h$	d_i	Sector	h	$\dim M_h$	d_i
0	0	1	1	d_1	6/13	7	$3d$
b	1	80	d^2	d_2	11/13	42	$3d$
a	1	81	s	d_3	15/13	119	$3d$
c_0	1	81	s	d_4	5/13	5	$3d$
c_1	1/3	3	s	d_5	7/13	13	$3d$
c_2	2/3	27	s	d_6	8/13	14	$3d$

There are two natural full CFTs built from U. One is the canonical diagonal theory of the completely rational net; the other is the $(E_8)_1$ theory of the holomorphic extension chosen below. Their partition functions are

$$Z_{\mathrm{diag}} = \sum_{i=0}^{11} |\chi_i|^2, \qquad Z_{E_8} = |\chi_{E_8}|^2 = |\chi_0 + \chi_b + 2\chi_a|^2. \tag{29}$$

The diagonal theory comes from the Longo–Rehren construction [28]. For the second, we transport the canonical Lagrangian algebra $I_{\mathcal{H}_3}(\mathbf{1})$ from $Z(\mathcal{H}_3)$ to $\mathrm{Rep}(U)$. It is a positive Q-system with trivial twist, so it gives a unitary holomorphic VOA extension H' [29, 25], which is of CFT type because the nonvacuum weights are positive. By the classification of holomorphic VOAs at $c = 8$, $H' \cong V_{E_8}$ [30]. Writing A_3 for the corresponding algebra in $\mathrm{Rep}(U)$, we have

$$\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3, \qquad {}_{A_3}\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}. \tag{30}$$

The opposite category accounts for the right action in $(x \boxtimes y) \cdot m = x \otimes m \otimes y$. These defects preserve U and the stress tensor, but need not commute with all of the E_8 currents. The first factor is the Haagerup symmetry.

We emphasize that H' need not be the original lattice extension H/U: for the latter we have only shown that the category of all right modules over its extension algebra in $\mathrm{Rep}(U)$ is equivalent to $\mathcal{H}_2$ or $\mathcal{H}_3$. Since $\chi_a = \chi_{c_0}$, the unflavoured partition function cannot tell the two apart. The diagonal theory and the E_8 theory, on the other hand, are easy to distinguish. In the diagonal theory the Verlinde lines form $Z(\mathcal{H}_3)$, and the lightest nontrivial scalar has dimension 2/3; in the E_8 theory the lightest nontrivial scalar has dimension 2.

Our construction thus realizes the Evans–Gannon character vector through explicit local fields and establishes Haagerup symmetry in the $(E_8)_1$ theory. It provides a concrete setting for studying how this symmetry acts on local operators and constrains their correlation functions.

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AI-use disclosure

AI systems were used to help with numerics and with preparation of this draft. The author takes responsibility for the accuracy and integrity of the final contents.

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